

Injecting Multimodal Information Into Pre-Trained Language Model for Multimodal Sentiment Analysis

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Abstract—With the increasing availability of computational and data resources, numerous powerful pre-trained language models (PLMs) have emerged for natural language processing tasks. However, how to inject nonverbal modalities into PLMs to handle multimodal information remains a practical problem. In this paper, we explore the application of PLM on multimodal sentiment analysis from a different perspective. Unlike many recent methods that develop multimodal fusion layers that are sequential to attention layers, we investigate the effectiveness of cross-modal additive attention that is parallel to attention layers, which takes the language modality as dominant modality. Moreover, we devise a gating mechanism to control the flow of nonverbal information by estimating its discriminative level. In this way, we can prevent noisy multimodal information from damaging the performance of pre-trained language model. In our framework, nonverbal modalities serve as auxiliary roles to provide the model with additional information and improve the understanding of multimodal human language. Additionally, cross-modal margin and matching losses are proposed to align the distributions of various modalities and simultaneously retain modality-specific information, which to some extent address the shortcoming of contrastive learning loss. Comprehensive experiments show that our approach surpasses existing state-of-the-art methods on multimodal sentiment analysis and emotion recognition tasks.

Index Terms—Multimodal sentiment analysis, multimodal representation learning, pre-trained language model, multimodal emotion recognition.

I. INTRODUCTION

MULTIMODAL machine learning is an important and emerging field in artificial intelligence, which has attracted significant research attention recently [1]. Specifically, a modality can be regarded as a unique source from which information is derived. In the real world, objects and events are often depicted through multiple modalities, and even humans perceive the world through various sensors (seeing, hearing, touching, etc). Multimodal sentiment analysis (MSA) [2],

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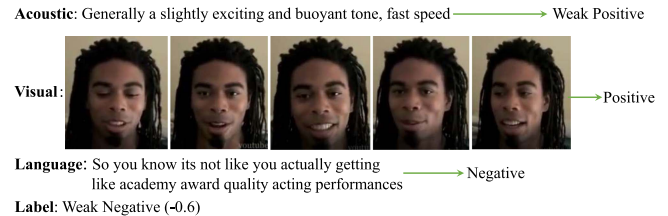


Fig. 1. A sample from the CMU-MOSI [4] dataset where visual and acoustic modalities contain noisy information.

[3], aiming to effectively integrate information from acoustic, spoken language, and visual expressions for the analysis of sentiment intensity, opinion tendency, and emotional state, is a promising area of multimodal machine learning and has great application potential.

The regular procedures of multimodal sentiment analysis mainly involve three steps: (1) learning unimodal representations via unimodal networks; (2) designing fusion approach to generate multimodal representation based on unimodal representations; (3) and inferring final prediction using the multimodal representation. The majority of prior works concentrate on developing modern fusion approaches to fully explore the interactions between multiple modalities [5], and popular fusion approaches include cross-modal attention [5], [6], [7], tensor fusion [8], [9], graph fusion [10], [11], [12], etc. Recently, with the rise of powerful pre-trained language models (PLMs) [13], [14], [15], a few methods try to fine-tune PLMs with multimodal information [13], [16], [17], [18]. For example, UniMSE [19] injects a multimodal fusion layer after the attention layer to incorporate nonverbal modalities into the T5 model [14]. Although effective, these PLM-based multimodal methods pay little attention to the discriminative level of nonverbal modalities. Existing methods often directly concatenate nonverbal features with language features at the input-level or devise fusion layers that are sequential to the attention layers, which can allow noisy or irrelevant acoustic and visual cues to degrade the performance of PLMs [18], [19], [20], [21]. Previous studies have revealed that language is the dominant modality for the MSA task [22], [23], while the other two modalities often contain noise and should serve as auxiliary roles. An example is shown in Fig. 1, where the pitch and facial expressions of the speaker convey sentiment information that is contradictory to the ground-truth sentiment. If we fail to appropriately regulate the flow of nonverbal information, those less discriminative or even noisy nonverbal cues might potentially diminish the capabilities of PLMs.

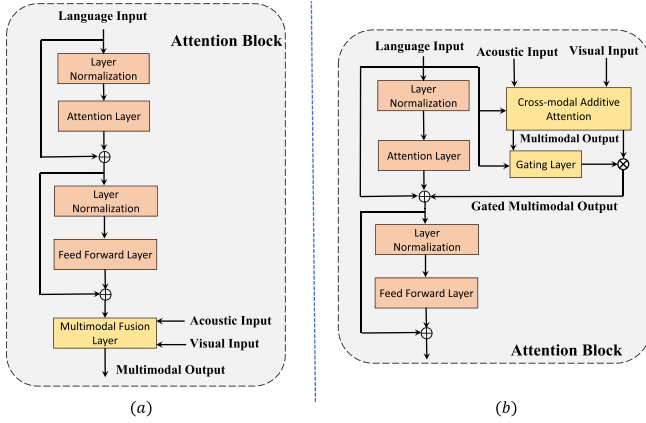


Fig. 2. The difference between (a) previous method [19] and (b) the proposed method in multimodal information modeling. We illustrate the difference using the encoder of T5 [14] as the backbone.

To address the above issue, we propose to inject nonverbal modalities into the PLM from a different perspective by introducing a parallel multimodal flow module with a gating mechanism that dynamically controls the influence of nonverbal modalities based on their estimated discriminative power. Specifically, to prevent the detrimental impact of noisy nonverbal information on the performance of PLM, we design a multimodal flow module which is parallel to the attention layer [24], making it easier to control the flow of nonverbal information compared to the sequential structure [18], [19] or the input-level concatenation of multimodal features [20], [21]. As shown in Fig. 2(b), multimodal flow module contains a cross-modal additive attention layer that generates a language-oriented multimodal representation, which takes language as the dominant modality. Moreover, we design a gating mechanism to control the flow of the generated multimodal representation according to its discriminative level. In this way, we can safeguard the discriminative power of PLM by avoiding the negative influence of noisy information in nonverbal modalities. It is because in the designed parallel multimodal flow module, we can easily control the influence of nonverbal features via the gating mechanism. If the nonverbal features are non-discriminative, the main pipeline of the model will not be seriously influenced as the gating values are relatively small. The parallel module enables the explicit decoupling of the language and nonverbal modalities, ensuring that the performance of the model is maintained even when only the language modality is effective. In contrast, as shown in Fig. 2(a), in the sequential structure where the multimodal fusion layer is injected before/after the attention layer (the input-level concatenation can be regarded as a special case of sequential structure), the output of the multimodal fusion layer is always considered; otherwise, there would be no signal to pass to the subsequent module. Consequently, it becomes difficult for sequential structure to explicitly avoid the negative impact of non-discriminative information in nonverbal modalities on the PLM. All in all, the proposed parallel multimodal flow module with a gating layer aims to effectively prevent noisy information in nonverbal modalities from interfering with the learning of human multimodal language. In addition, nonverbal modalities

serve as auxiliary roles to provide additional information to the powerful PLM and improve the model's understanding of human multimodal language.

Furthermore, since the pre-trained transformer model only accepts language modality during the pre-training stage, it cannot well recognize the information hidden in the nonverbal features. To this end, we propose to reduce the distributional gap between the language and nonverbal modalities. Contrastive learning is a common way for alignment in many previous methods [19], [25], [26], [27]. But it tends to explicitly encourage unimodal representations from the same multimodal sample to be as similar as possible, which ignores the existence of modality-specific information and may lead to information loss [28]. In contrast, we propose a cross-sample matching loss to align the expectation and variance of each feature between different modalities, which avoids the direct alignment of unimodal representations from the same sample. To be more specific, we calculate the mean and variance of each feature for each modality across samples in the mini-batch, after which we match the mean and variance of each nonverbal feature to that of the corresponding language one. This simple practice is shown to be effective in our experiment. Moreover, apart from aligning features from a cross-sample perspective, inspired by the triplet loss [29], we design the margin loss to align cross-modal features with a margin in a sample-specific manner. Different from contrastive learning that rigorously aligns the features between language and nonverbal modalities belonging to the same sample, the proposed margin loss requires the mutual information between the anchor and its positive to be larger than that between the anchor and its negative by a margin. In this way, cross-modal alignment can be achieved and meanwhile the modality-specific information of nonverbal modalities can be preserved. Combining cross-sample matching loss and sample-specific margin loss, we can align the distributions between multiple modalities in a more comprehensive manner. Notably, during the alignment, we stop the gradient flow to the language modality and only change the distributions of nonverbal modalities. The designed cross-modal margin loss and matching loss align the distributions of nonverbal modalities to that of the language modality. By better aligning the distributions, PLM is able to interpret and understand nonverbal features more accurately such that the model can effectively fuse the features from multiple modalities, which improves the model's understanding of human multimodal language.

To sum up, we introduce a novel architecture named multimodal gating transformer (MGT) to analyze human multimodal sentiment. The key contributions of MGT are outlined below:

- We propose a multimodal framework that explores a different solution to inject nonverbal modalities into PLM for inferring human sentiment. Different from previous methods that conduct regular cross-modal attention or apply sequential multimodal fusion layer [16], [18], [19], we design a multimodal flow module that is parallel to attention layer, which prevents the less-discriminative nonverbal modalities from diminishing the performance of PLM.
- In the multimodal flow module, drawing inspiration from the observations that language is the dominant modality in sentiment analysis and nonverbal modalities often contain

noise, we design cross-modal additive attention to generate language-focused multimodal representation, ensuring the dominant role of the language modality. Moreover, to control the flow of nonverbal information, we design a gating mechanism to identify the discriminative level of multimodal representation.

- We propose to align the distributions of nonverbal modalities to that of the language modality via the cross-modal margin and matching losses, which can maintain the unique information within each modality during modality alignment and thus help to generate a more informative multimodal representation for sentiment prediction.
- Extensive experiments suggest that MGT with T5 [14] as the PLM reaches state-of-the-art performance on the task of MSA. Moreover, we additionally evaluate MGT on multimodal emotion recognition (MER) and the results show that MGT also reaches very competitive performance.

II. RELATED WORK

A. Multimodal Sentiment Analysis

Multimodal sentiment analysis has received significant attention in the past few years. Previous works mainly focus on developing effective fusion methods to model cross-modal interactions [3], [30], [31], designing robust algorithms to handle noisy modalities [32], [33], proposing weakly-supervised and self-supervised multimodal learning methods to deal with the problem of limited training samples [25], [28], devising effective algorithms to address modality gap and regularize the learning of unimodal distributions [34], [35], [36], etc. The methods that focus on designing attention-based fusion approaches are closely related to our MGT. Specifically, applying attention mechanisms to explore cross-modal interactions is a popular approach, which focuses on exploring highly similar cross-modal context information and paying attention to the more important information in multimodal data [6], [7], [37]. For example, Sentiment Knowledge Enhanced Attention Fusion Network (SKEAFN) [38] adds additional sentiment knowledge representation from an external knowledge base to language modality, and designs a feature-wised attention fusion module to dynamically adjust the weights of sentiment knowledge representation and unimodal representations for multimodal fusion, so as to enhance the effect of fusion. Different from these attention mechanisms, the proposed cross-modal additive attention modifies the traditional self-attention in Transformer by adding the prior knowledge that language is the dominant modality to generate language-focused multimodal representation and reduce the noise from nonverbal modalities. Moreover, these methods do not pay attention to how to fine-tune PLMs and leverage the power of PLMs to improve the understanding of human multimodal language.

In particular samples, the information carried by specific modalities might not have discriminative power, and noise is inevitably introduced in the process of data acquisition and data processing, leading to the noisy modality problem. To resolve this issue, some fusion methods have been proposed. Chen et al. [39] use reinforcement learning to train a gate controller for each modality, which can explicitly filter out noisy

information at each time slice. However, reinforcement learning algorithms introduce additional parameters and optimization objectives [40], which easily leads to training instability. Recently, to filter out noisy information, Zeng et al. [41] propose soft/hard mask mechanisms to identify the contributions of different modalities and thus recognize noisy modalities. Compared with using reinforcement learning to obtain exact binary weights [42], the hard mask mechanism proposed by Zeng et al. [41] can explicitly calculate the gradients of parameters, which is simple to operate and stable to train. Temporal Tensor Fusion Network (T2FN) [43] handles the noisy modality problem and missing data from another perspective. T2FN considers clean high-dimensional multimodal sequential tensors to be low-rank, while the presence of noisy information and missing data cause the low-rank tensors to become high-rank. Therefore, T2FN uses the tensor rank minimization constraint to learn the representations of tensors, achieving the effect of filtering noisy information and improving the quality of tensors. More recently, Multi-modal Boosting [44] designs a weakly-supervised algorithm to explicitly learn the discriminative level of various modalities via the supervision from unimodal losses, and thus identifies noisy modalities. However, these methods do not take into account the prior knowledge that nonverbal modalities contain more noise to filter out the noise in nonverbal modalities effectively, which is different from our MGT.

Additionally, to avoid the negative impact of modality gap, modality-translation methods [5], [22], [45] reduce the modality gap by translating the features of one modality into the features of another modality. Furthermore, some approaches leverage Kullback–Leibler divergence (KL-divergence) to regularize the learning of unimodal distributions and reduce the gap between multiple modalities [46]. For instance, Sun et al. [35] apply KL-divergence and propose a meta-learning algorithm to learn favorable unimodal representations for multimodal tasks. Inspired by the Generative Adversarial Network (GAN) [47], Adversarial Representation Graph Fusion (ARGF) [10] adopts adversarial learning to learn a common feature space and reduce the distribution difference between modalities in the feature space. However, the training of GAN often suffers from instability. In addition, Modality-Invariant and -Specific representations (MISA) [34] maps the features of each modality into two subspaces, learning representations that are invariant across modalities and are specific to individual modalities. By applying orthogonal constraint and central moment discrepancy constraint to these representations, MISA can effectively reduce the distribution gap between modalities and decouple unimodal features. However, MISA requires additional resources to preserve features from multiple subspaces. Yang et al. [48] combine the characteristics of ARGF and MISA, using GAN to learn representations that are shared across modalities and specific to individual modalities. Yang et al. propose a modality discriminator to learn common and private encoders for individual modalities using adversarial training, and uses consistency loss to constrain the shared representations of all the modalities to be consistent. In contrast to these methods, we propose the cross-modal margin and matching losses to match the distributions between nonverbal modalities and language modality, which

achieve the modality alignment from both sample-specific and cross-sample perspectives in a comprehensive manner.

Moreover, to further reduce the modality gap and heterogeneity between various modalities, contrastive learning is employed to pull closer the unimodal representations from the same sample in the embedding space [25], [26], [27], [49], [50]. For example, Fan et al. [27] propose a multi-level contrastive learning strategy to alleviate the heterogeneity between multiple modalities and mitigate the heterogeneity between unimodal and multimodal representations. However, contrastive learning often encourages the unimodal representations from the same multimodal sample to be identical, resulting in the loss of modality-specific information hidden in each modality [28]. In contrast, the proposed cross-modal matching and margin losses aim to preserve the modality-specific information while aligning various modalities, such that the generated multimodal representation can be more informative.

B. Fine-Tuning PLMs With Multimodal Data

In the past few years, with the rise of BERT [13], a resurgence of work of fine-tuning pre-trained models with multimodal information has led to major advances for its remarkable ability in understanding multimodal data with limited training samples [18], [51], [52], [53], [54]. For instance, Cross-modal BERT (CM-BERT) [16] designs a cross-modal attention layer to enable BERT to learn from acoustic features. Rahman et al. [18] inject the multimodal adaptation gate [55] into BERT [13] and XLNet [56] such that they can process multimodal information during fine-tuning. All-modalities-in-One BERT (AOBERT) [17] proposes multimodal masked language modeling and alignment prediction to further pretrain BERT to sufficiently explore intra- and inter-modalities correlation. Moreover, Liu et al. [57] use advanced ensemble techniques to fully take advantage of multiple pre-trained models (e.g., GPT-2 [58] and BERT [13]) for language and visual modalities, demonstrating the synergistic potential of combining different modal analytical strengths. However, they pay little attention to the modality gap problem and treat the two modalities as equal. Prompt learning is a popular technique that is able to inject multimodal features into PLMs to activate the capability of language models for understanding multimodal information with limited samples [59], [60], [61], [62]. For example, Context Optimization (CoOp) [63] is a very classical work that introduces the concept of prompt learning to the vision domain for adapting pre-trained vision-language models, modeling a prompt's context words with learnable vectors while the entire pre-trained parameters are kept fixed. Yu et al. [64] introduce multimodal prompt learning to sentiment analysis, proposing a unified pre-training framework for multimodal prompt-based fine-tuning to narrow the semantic gap between language and vision. Furthermore, Yang et al. [65] propose a unified multimodal prompt framework composed of image description prompts and text prompts, utilizing a probability fusion module to combine multiple predictions from different prompts. However, they usually directly concatenate visual features with language features. In contrast, we design a parallel multimodal flow module to process nonverbal features to explicitly control

the flow of multimodal information. Recently, UniMSE [19] proposes a sequential multimodal fusion layer to incorporate visual and acoustic information into T5 [14] and uniformly formulates MSA and MER as generation tasks. Different from it, to avoid diminishing the performance of PLM by directly injecting nonverbal modalities, we design a multimodal flow module that is parallel to self-attention layer, and design a gating mechanism to control the flow of nonverbal information according to their discriminative level. In this way, we can guarantee that the power of PLM will not be influenced by noisy nonverbal information. Furthermore, we propose margin loss and matching loss to reduce the distributional gap between modalities and simultaneously preserve the modality-specific information, which is often ignored in previous works.

III. ALGORITHM

The diagram of MGT is shown in Fig. 3, and the detailed introduction of MGT is presented in the following sections. The main downstream task of this paper is MSA. To demonstrate the generalization ability of MGT, we additionally evaluate MGT on the MER task. MSA is a regression task focused on estimating the sentiment intensity expressed by humans, whereas MER is concerned with identifying the type of emotion being conveyed by humans in a video utterance. MGT processes video utterances, which are characterized by three modalities, i.e., language (l), acoustic (a), and visual (v). Following the feature extraction process detailed in Section IV-C, acoustic, visual, and language modalities are encapsulated in feature sequences $U_a \in \mathbb{R}^{T_a \times d_a}$, $U_v \in \mathbb{R}^{T_v \times d_v}$, and $U_l \in \mathbb{R}^{T_l \times d_l}$ respectively, where T_m denotes the sequence length and d_m is the feature dimensionality for individual modalities (with m belonging to the modality set $\mathcal{M} = \{a, v, l\}$).

A. Unimodal Networks

Formally, we have three modalities as input: $U_l \in \mathbb{R}^{T_l \times d_l}$, $U_v \in \mathbb{R}^{T_v \times d_v}$, $U_a \in \mathbb{R}^{T_a \times d_a}$. We first introduce how we obtain the latent features of visual and acoustic modalities. For the unimodal networks of acoustic and visual modalities, following previous works [19], [66], we use a simple LSTM network for each modality as the unimodal network:

$$X_m = \text{LSTM}_m(U_m; \theta_m) \in \mathbb{R}^{T_m \times d_m} \quad (1)$$

where $X_m \in \mathbb{R}^{T_m \times d_m}$ is the processed unimodal sequence for modality m . We take the last hidden state of X_m as the final unimodal representation $x_m \in \mathbb{R}^{d_m}$, which is used for multimodal fusion. Since for the MSA task, visual and acoustic modalities often contain noisy information that interferes the learning of the model, here we do not use visual and acoustic sequences for fusion. Instead, we compress the information from nonverbal modalities and ensure the dominant role of language information, which also helps to reduce the time complexity of the model. For the language modality, following state-of-the-art method [19], we use the pre-trained tokenizer of T5 [14] to encode the language input before feeding it to MGT.

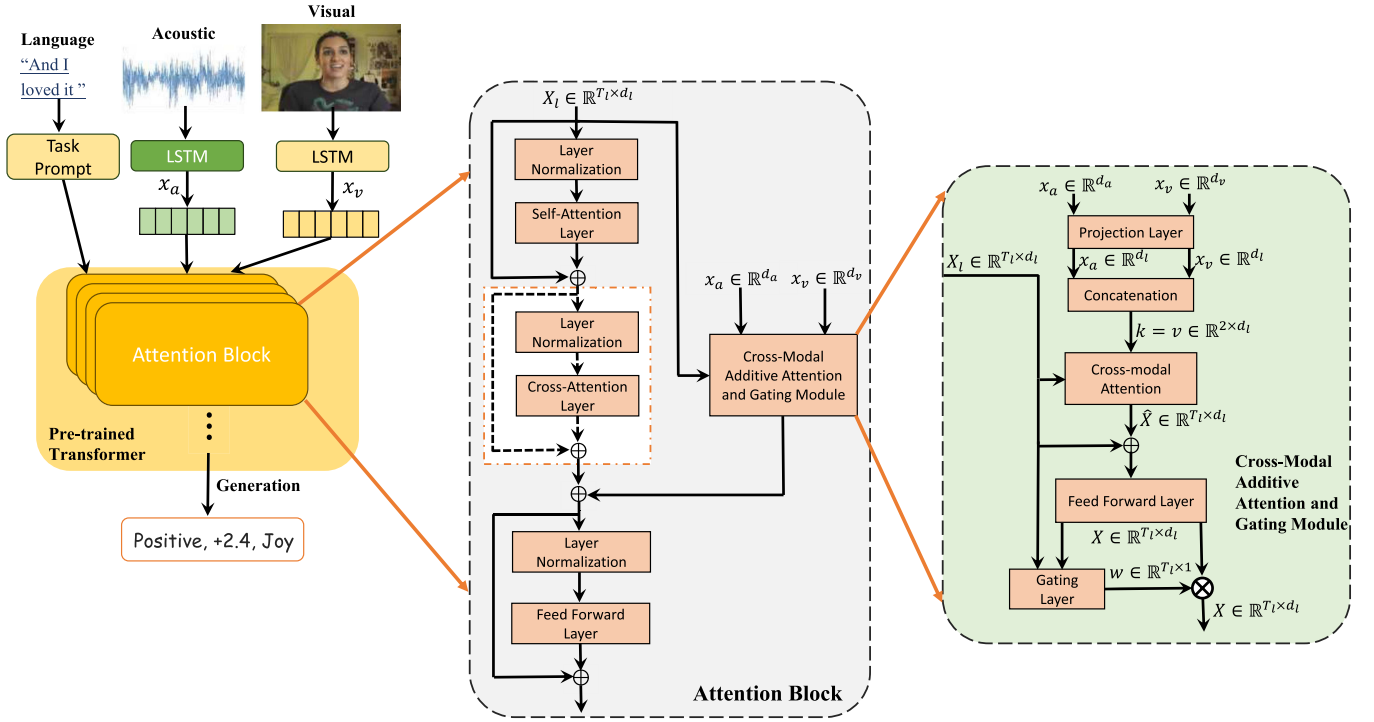


Fig. 3. The diagram of the proposed MGT. We use T5 [14] as our pre-trained language model, and the module within the dashed boxes only applies to the decoder of T5.

B. Multimodal Flow Module

The proposed multimodal flow module consists of a cross-modal additive attention layer and a gating layer. To be more specific, in contrast to previous methods that directly concatenate nonverbal features with the language tokens or design a multimodal fusion layer that is sequential to the attention layer [16], [18], [19], in the multimodal flow module, we propose to devise a cross-modal additive attention layer that is parallel to the attention layer. The cross-modal additive attention is designed based on the observations that language is the dominated modality for inferring human sentiment and nonverbal modalities often contain noisy information, which can effectively acquire the language-focused multimodal representation and prevent nonverbal features from damaging the performance of the powerful PLM. In addition, we design a gating layer to further control the flow of nonverbal information according to their estimated discriminative level, ensuring that non-discriminative multimodal information will not diminish the performance of PLM.

1) *Cross-Modal Additive Attention*: In this section, we describe the proposed cross-modal additive attention layer in detail. The cross-modal additive attention layer takes the language representation as the query, and take nonverbal features as the key and value. By conducting cross-modal additive attention, we can acquire language-attended nonverbal features. Then, we add the language features with the attended nonverbal features to obtain the multimodal representation that is dominated by language modality, such that it can be well recognized by the pre-trained language model and prevent the language model from being damaged by noisy features of nonverbal modalities.

Specifically, after we obtain the representations of visual and acoustic modalities output by unimodal networks, we first map the feature dimensionality of visual and acoustic features into the same dimensionality as that of the language representation, and then concatenate the visual and acoustic features at the time dimension:

$$x_m \leftarrow x_m W_m^{map} + b_m^{map} \in \mathbb{R}^{d_l} \quad (2)$$

$$k = v = \text{unsqueeze}(x_a) \oplus \text{unsqueeze}(x_v) \in \mathbb{R}^{2 \times d_l} \quad (3)$$

where $m \in \{a, v\}$, W_m^{map} is the learnable mapping parameter matrix and b_m^{map} is the learnable bias vector, k and $v \in \mathbb{R}^{2 \times d_l}$ denote the key and query embedding for cross-modal attention. In the *unsqueeze* operation, we expand the dimension of the nonverbal features (the new dimension is called time dimension), and $\oplus$ denotes the concatenation. Notably, the time dimensionality of k and v is merely two and is significantly smaller than T_l , which to some extent compresses the information from nonverbal modalities and ensures the dominant role of language information.

Then, the cross-modal additive attention layer takes the language representation $X_l \in \mathbb{R}^{T_l \times d_l}$ as the query, and conducts regular cross-modal attention as follows:

$$\hat{X} = \text{softmax}\left(\frac{X_l k^T}{\sqrt{d_l}}\right) v \quad (4)$$

where T is the matrix transpose operation. In this way, the output $\hat{X}$ denotes the weighted sum of visual and acoustic features where the weights are determined by language features. However, $\hat{X}$ has not yet incorporated the features of the language

modality. Therefore, different from previous cross-modal attention method [54], we further add the language representation to $\hat{\mathbf{X}}$ to obtain language-oriented multimodal representation, and apply a feed forward network to process the multimodal representation:

$$\mathbf{X} = FFN(\mathbf{X}_l + \hat{\mathbf{X}}) \quad (5)$$

where the feed forward network FFN contains two $ReLU$ activated fully connected layers that first map the input into a low-dimensional feature space and then restore the feature dimensionality to the original one. The language-oriented multimodal representation $\mathbf{X}$ incorporates the information from three modalities, which can enhance the ability of PLM to understand human multimodal language.

2) *Gating Layer*: The discriminative level of the multimodal representation may vary from samples to samples, and even some of the multimodal information is noisy. Considering that the pre-trained language model does not have access to the acoustic and visual information during pre-training, it may fail to well identify the discriminative level of multimodal information and thus is easily affected by noisy information, which might result in performance degradation.

To address this issue, we further design the gating layer to control the flow of multimodal information. The equation is shown as follows:

$$\mathbf{w} = \text{Gating}(\mathbf{X}_l, \mathbf{X}) = \sigma((\mathbf{X}_l \oplus \mathbf{X})\mathbf{W}_g + \mathbf{b}_g) \quad (6)$$

$$\mathbf{X} \leftarrow \mathbf{X} * \mathbf{w} \quad (7)$$

where $\mathbf{w} \in \mathbb{R}^{T_l \times 1}$ is the learned weight for multimodal representation, $\mathbf{W}_g$ and $\mathbf{b}_g$ are the learnable parameters for the gating layer, $\oplus$ is the concatenation operation at the feature dimension, and σ is the Sigmoid activation function. The gating layer takes language representation and multimodal representation as inputs to estimate the effectiveness and similarity of the multimodal representation with respect to the language representation, and outputs a scalar that denotes the discriminative level of the multimodal representation. The weight $\mathbf{w}$ is further multiplied with the multimodal representation to generate the gated multimodal representation. Finally, the gated multimodal representation is added to the original language representation to enhance the capability of the model to understand multimodal human language:

$$\mathbf{X}_l \leftarrow \mathbf{X}_l + \mathbf{X} \quad (8)$$

The obtained $\mathbf{X}_l$ will be passed into the next attention block. And finally, the model is encouraged to generate the desired output sequence of the MSA and MER tasks for each specific sample based on $\mathbf{X}_l$ output by the final attention block.

C. Distribution Alignment

The PLM only accepts language modality as input at the pre-trained stage, and the language modality is proved to be the dominant modality for inferring human sentiment [22], [23]. Therefore, it is necessary to align the distributions between nonverbal modalities and the language modality, which can

enable the PLM to better understand nonverbal features and improve the performance of MGT. To this end, we introduce the proposed matching loss and margin loss which align the distributions of nonverbal modalities to that of the language modality. Contrastive learning is a widely-used approach for aligning different modalities [25], [26], [27], [67], [68]. However, it tends to rigorously force the features between different modalities of each sample to be identical, potentially leading to the loss of information unique to each modality and resulting in the degradation of multimodal representation [28]. In contrast, we propose the matching loss and margin loss to align the distributions of various modalities and avoid the explicit alignment of unimodal representations from the same sample, which can simultaneously preserve the modality-specific information. The matching loss and margin loss achieve distribution alignment from cross-sample and sample-specific manners respectively, leading to a more comprehensive alignment.

1) *Matching Loss*: Matching loss reduces the modality gap between nonverbal modalities and language modality via a cross-sample perspective. Specifically, taking acoustic modality as an example, given the language sequence $\mathbf{X}_l$ and acoustic sequence $\mathbf{X}_a$, we first calculate the compressed language representation $\hat{\mathbf{x}}_l$ and acoustic representation $\hat{\mathbf{x}}_a$ via average pooling at the time dimension:

$$\hat{\mathbf{x}}_l = \text{pooling}(\mathbf{X}_l \in \mathbb{R}^{T_l \times d_l}) \in \mathbb{R}^{d_l}$$

$$\hat{\mathbf{x}}_a = \text{pooling}(\mathbf{X}_a \in \mathbb{R}^{T_a \times d_a}) \in \mathbb{R}^{d_a} \quad (9)$$

where *pooling* denotes the pooling operation that calculates the average of the input sequence along the time dimension. To map the feature dimensionality of acoustic modality to the same as that of language modality, we have the following equation:

$$\hat{\mathbf{x}}_a \leftarrow \hat{\mathbf{x}}_a \mathbf{W}_a + \mathbf{b}_a \in \mathbb{R}^{d_l} \quad (10)$$

where $\mathbf{W}_a \in \mathbb{R}^{d_a \times d_l}$ is a learnable mapping matrix and $\mathbf{b}_a$ is the learnable bias vector. The mapping operation projects nonverbal features into a metric space where they can be directly compared with language representations in terms of similarity. Afterwards, we calculate the mean (expectation) of each feature in each modality along the batch dimension as follows:

$$\text{mean}_i^m = \frac{1}{n} \sum_{j=1}^n (\hat{x}_m^j)_i, \quad i \in \{1, 2, \dots, d_l\} \quad (11)$$

where $(\hat{x}_m^j)_i$ denotes the i^{th} feature of modality m of sample j ($m \in \{l, a, v\}$), n is the batch size, and mean_i^m is the average of feature i in modality m ($m \in \{l, a, v\}$). The variance of each feature in each modality along the batch dimension is calculated as follows:

$$\text{var}_i^m = \frac{1}{n} \sum_{j=1}^n ((\hat{x}_m^j)_i - \text{mean}_i^m)^2 \quad (12)$$

where var_i^m is the variance of feature i in modality m . Then, we match the mean and variance of each feature in acoustic modality to the corresponding one in language modality:

$$L_M^a = \frac{1}{d_l} \sum_{i=1}^{d_l} ((\text{mean}_i^a - \text{mean}_i^l)^2 + (\text{var}_i^a - \text{var}_i^l)^2) \quad (13)$$

where L_M^a is the matching loss for acoustic modality, and L_M^v is calculated similarly. The total cross-modal matching loss is calculated as:

$$L_M = L_M^a + L_M^v \quad (14)$$

In this way, the cross-modal matching loss L_M matches the distributions of nonverbal modalities and language modality via a cross-sample perspective. In other words, since $mean^m$ and var^m are calculated based on a batch of samples, the optimization guided by L_M can avoid the explicit alignment of cross-modal features within each sample, which helps to preserve modality-specific information within each modality and is difficult to be achieved by contrastive learning [19], [25]. The retaining of modality-specific information helps to generate a more informative multimodal representation for inferring the sentiment more accurately.

2) *Margin Loss*: Different from cross-sample matching loss, the margin loss aligns the features of various modalities from a sample-specific perspective. Nevertheless, margin loss also tries to preserve the modality-specific information via the introduced margin. The proposed margin loss is inspired by the triplet loss [29]. In this margin loss, the anchor is set to the representation of the language modality, while the positive of the anchor is the corresponding acoustic or visual representation from the same sample. Unlike the traditional triplet loss, the negative in the margin loss is defined as the average of the acoustic/visual representations from all other samples in the batch. In this way, we can take advantage of many negatives for each anchor to conduct modality alignment in a more effective and robust way [25].

Specifically, similar to the mapping operation in the matching loss (10), we first map the feature dimensionality of acoustic and visual modalities into that of language modality using the same mapping matrix. Then, we normalize the representations of anchor, positive and negative via the L2 normalization respectively:

$$(\hat{x}_m^j)_i \leftarrow \frac{(\hat{x}_m^j)_i}{\sqrt{\sum_{g=1}^{d_l} (\hat{x}_m^j)_g^2}}, \quad i \in \{1, \dots, d_l\} \quad (15)$$

where the superscript j ($j \in \{1, \dots, n\}$) denotes the index along the batch dimension, and subscript m ($m \in \{l, a, v\}$) indicates the specific modality, i and g denote the positions along the feature dimension. L2 normalization is applied to keep the similarity values between unimodal representations within the range of -1 to 1. Then, the negative for the anchor $\hat{x}_l^j$ is defined as:

$$neg_m^j = \frac{1}{n-1} \sum_{b \neq j} \hat{x}_m^b, \quad b \in \{1, \dots, n\}, \quad b \neq j \quad (16)$$

where $m \in \{a, v\}$, b is the index at the batch dimension, and neg_m^j is the negative for the anchor $\hat{x}_l^j$. Finally, we use the dot product to calculate the similarity between anchor and its positive/negative, and define the cross-modal margin loss as

follows:

$$L_{Margin}^m = \frac{1}{n} \sum_{j=1}^n ReLU(\hat{x}_l^j (neg_m^j)^T - \hat{x}_l^j (\hat{x}_m^j)^T + \gamma) \quad (17)$$

$$L_{Margin} = \sum_{m \in \{a, v\}} L_{Margin}^m \quad (18)$$

where L_{Margin} denotes the margin loss, $ReLU$ is the activation function satisfying $ReLU(x) = \max(0, x)$, $\hat{x}_m^j$ is the positive for the anchor $\hat{x}_l^j$, T represents matrix transpose operation, and γ is the margin which is a scalar between 0 and 1. Notably, for margin and matching losses, we stop the gradients to the language representation and only change the distributions of nonverbal modalities.

Different from contrastive learning that explicitly aligns the representations between different modalities belonging to the same sample, margin loss requires the mutual information between the anchor and its positive to be larger than that between the anchor and its negative with a margin. In this way, margin loss aligns the features of various modalities belonging to the same sample, and at the meantime it is capable of preserving the modality-specific information of nonverbal modalities.

The visualization of the designed cross-modal matching loss and margin loss is shown in Fig. 4. Combining the cross-sample matching loss and sample-specific margin loss, we can align the distributions between multiple modalities in a more comprehensive way.

D. Task Prompt

We design detailed task-aware prompt for each task to guide the token generation of the PLM, reducing the difficulty of model learning and task comprehension. The designed prompt is concatenated with the input spoken language. Specifically, for the task of MSA, the prompt is set as ‘Predict the sentiment (positive, neutral, negative) and sentiment intensity (a number ranging from -3 to 3, larger number indicates more positive sentiment) of the input: ’. For MER, the prompt is designed as ‘Predict the emotion (choose from joy, sadness, fear, anger, surprise, disgust, neutral) of the input: ’. For MSA, the target output sequence is formulated as: ‘[sentiment], [sentiment intensity]’ where ‘sentiment intensity’ is a value and ‘sentiment’ is chosen from positive, neutral and negative. For MER, the target output sequence is formulated as: ‘Emotion is [xx]’ where ‘xx’ is chosen from the above seven kinds of emotions. Notably, all the datasets are trained jointly to perform multi-task training.

E. Prediction and Loss Calculation

Different from the traditional methods that formulate MSA and MER tasks as discriminative tasks, the proposed MGT formulates MSA and MER tasks as generation tasks and forces the model to generate the desired sequences for these tasks [19]. Then, we apply the cross-entropy loss to calculate the generation loss as the task loss. Finally, we add the cross-modal matching

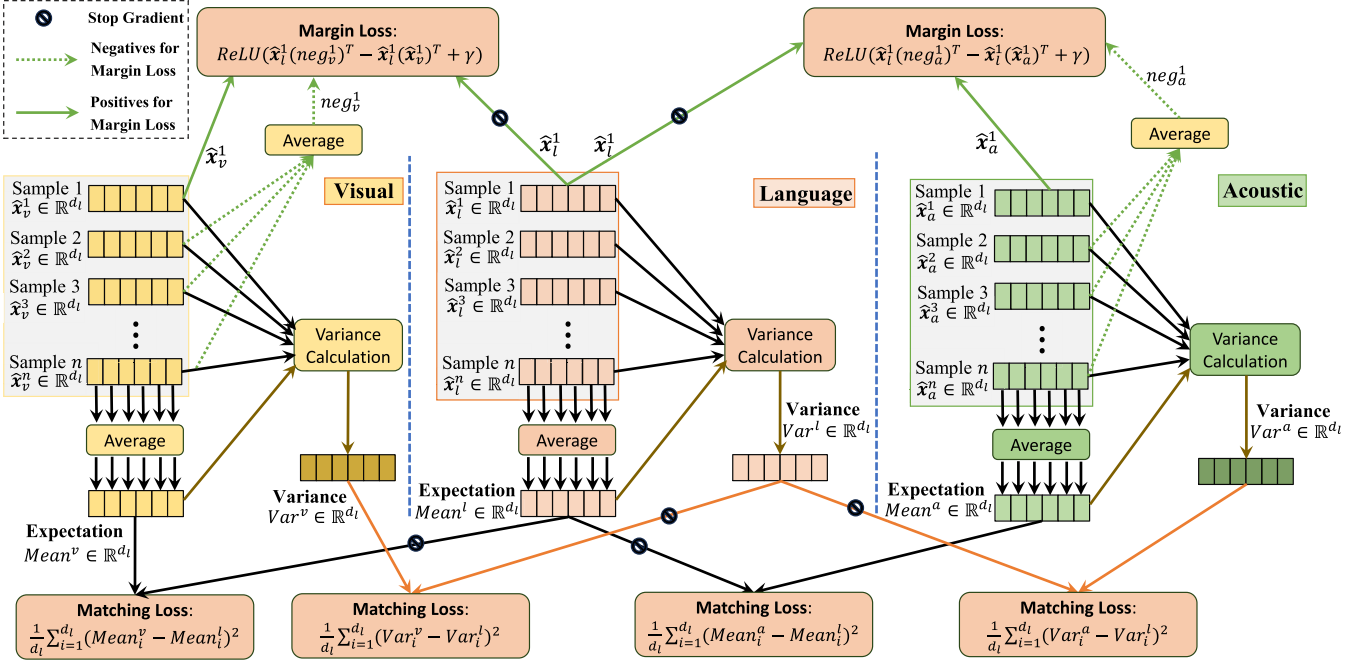


Fig. 4. Visualizations of matching and margin losses. We only illustrate the margin loss for Sample 1 for clarify.

loss and margin loss to obtain the total loss of the proposed framework.

Specifically, we extract the label token (sentiment, sentiment intensity, emotion) from the output sequence of the model, and apply the cross-entropy loss to compute the downstream task loss using the label token:

$$\mathcal{L}_{task} = -\frac{1}{n} \sum_{j=1}^n \sum_i y_i^j \log \hat{y}_i^j \quad (19)$$

where y^j is the ground truth one-hot label vector for sample j , and $\hat{y}_i^j$ is the predictive probability for the i^{th} token in the vocabulary of the T5 tokenizer. The final loss is defined as:

$$\mathcal{L} = \mathcal{L}_{task} + \alpha \cdot (\mathcal{L}_{Margin} + \mathcal{L}_M) \quad (20)$$

where α is the weight for the margin and matching losses.

IV. EXPERIMENTS

In this section, we evaluate the proposed MGT on MSA and MER tasks across three datasets.

A. Datasets

(1) *CMU-MOSI* [4] is a commonly-used dataset in the field of MSA, which is composed of more than 2,000 video utterances. Each video utterance is annotated with sentiment intensity on a Likert scale ranging from -3 to 3, with 3 indicating the most intense positive sentiment and -3 indicating the most intense negative sentiment. To be consistent with previous algorithms [19], [66], our algorithm uses 1,284 utterances as training set, 229 utterances as validation set, and 686 utterances as testing set.

(2) *CMU-MOSEI* [11] is a large scale dataset of MSA and MER. The CMU-MOSEI dataset is comprised of over 20,000

video utterances sourced from over 1,000 speakers on YouTube, encompassing 250 unique topics. These utterances are selected at random from a variety of topics and monologue videos. We leverage the sentiment labels of CMU-MOSEI to perform MSA task, which have the same range as those in the CMU-MOSI dataset. To be consistent with prior methods [66], [69], we employ a training set of 16,326 utterances, a validation set of 1,871 utterances, and a testing set of 4,659 utterances.

(3) *MELD* [70] is a widely-used dataset for MER collected from a show called Friends. MELD is composed of 13,707 video utterances of conversations between multiple persons, and involves seven universal emotions, including joy, sadness, fear, anger, surprise, neutral, and disgust.

B. Compared Baselines

We compare our MGT with multiple competitive baselines, including *Self-MM* [66], *TFN* [8], *LMF* [71], *MMIM* [69], *MISA* [34], *UniMSE* [19], *ConFEDE* [49], *ITHP* [36], *DMD* [72], *MM-DFN* [73], *MM-GCN* [74], *DAG-ERC* [75], *DialogueGCN* [76], where UniMSE is the most similar to our approach. The detailed introduction is as follows:

(1) *Self-Supervised Multi-task Multimodal sentiment analysis network (Self-MM)* [66]: Self-MM obtains sentiment labels for each modality by utilizing the annotated global labels of multimodal samples to learn meaningful and discriminative representations for each modality; (2) *Tensor Fusion Network (TFN)* [8]: TFN computes the outer product of the three unimodal representations to generate expressive multimodal tensor that explicitly captures interactions at the uni-modal, bi-modal, and tri-modal levels; (3) *Low-rank Modality Fusion (LMF)* [71]: LMF decomposes the weight tensors of the generated multimodal tensor into low-rank tensors,

which can reduce the space and time complexity of tensor fusion; (4) MultiModal InfoMax (MMIM) [69]: MMIM simultaneously maximizes the mutual information among various unimodal representations and between the multimodal and unimodal representations, such that more expressive multimodal representations can be learned; (5) Modality-Invariant and -Specific Representation (MISA) [34]: MISA projects each unimodal representation into modality-specific and modality-invariant subspaces and then reduces the modality gap; (6) Unified MSA and ERC (UniMSE) [19]: UniMSE unifies the multimodal sentiment analysis and multimodal emotion recognition tasks and formulates them as generation tasks; (7) Contrastive FEature DEcomposition (ConFEDE) [49]: ConFEDE concurrently executes contrastive representation learning and contrastive feature decomposition, thereby enhancing the multimodal representation; (8) Contrastive Learning Guided by Sentiment Intensity (CLGSI) [26]: CLGSI selects and highlights positive and negative sample pairs of contrastive learning based on the difference in sentiment intensity and tries to extract common features between various modalities; (9) Decoupled Multimodal Distillation (DMD) [72]: To enhance the discriminative features of each modality, DMD facilitates flexible and adaptive cross-modal knowledge distillation, which decouples each unimodal representation into modality-irrelevant/-exclusive spaces and employs a graph distillation unit to process each decoupled part in a more specialized and effective manner; (10) Information-Theoretic Hierarchical Perception (ITHP) [36]: Based on information bottleneck, ITHP designates a prime modality and regards the remaining modalities as detectors in the information pathway that serve to distill the flow of information; (11) Multi-level Contrastive Learning and Multi-layer Convolution Fusion ((MCL-MCF) [27]: MCL-MCF proposes a multi-level contrastive learning strategy to alleviate the heterogeneity between multiple modalities and mitigate the heterogeneity between unimodal and multimodal representations; (12) Multimodal Dynamic Fusion Network (MM-DFN) [73]: MM-DFN proposes a graph-based fusion network to fuse multimodal context features in conversation; (13) MultiModal fused Graph Convolutional Network (MM-GCN) [74]: MM-GCN utilizes the speaker information to model inter-speaker and intra-speaker dependency; (14) Directed Acyclic Graph for Emotion Recognition from Conversation (DAG-ERC) [75]: DAG-ERC combines the advantages of GCNs and recurrence-based neural networks to capture the flow of information between the broader conversational context and nearby context; (15) Dialogue Graph Convolutional Network (DialogueGCN) [76]: DialogueGCN proposes a GCN to capture contexts of emotions in human conversations.

C. Feature Extraction

We utilize the following tools and models to extract the features of acoustic, visual, and language modalities.

(1) *Visual modality*: Following the state-of-the-art model [19], we extract T_v frames from each video utterance and utilize the effcientNet [77] that is pre-trained on VGGface and AFEW

dataset to derive the visual features. The sequence length of the visual features is 32. (2) *Acoustic modality*: We process raw acoustic signal into sequential vectors using librosa,¹ which extracts Mel-spectrogram to form a sequence of acoustic features. The Mel-spectrogram represents the short-term power spectrum of the acoustic signal, which is popular in audio processing. The sequence length of the acoustic features is 157. (3) *Language modality*: Following state-of-the-art method [19], T5-Base [14] is used as the backbone of the language modality. In each mini-batch, we use padding to ensure all the language sequences to have the same length, and the length varies across different mini-batches.

Following UniMSE [19], since the contextual information is of vital importance to language understanding, for language modality, we concatenate the text from current utterance with the text from its former 2-turn and latter 2-turn utterances as the final input raw text. Additionally, we set segment id to distinguish the current utterance and its contexts. For more details, please refer to [19].

For the MELD dataset, the respective dimensionality of the language, acoustic, and visual features are 768, 60, and 36. For the CMU-MOSI dataset, the dimensionality of the corresponding unimodal features are 768, 74, and 47, respectively. For the CMU-MOSEI dataset, the dimensionality of the corresponding unimodal features are 768, 74, and 35, respectively.

D. Evaluation Metrics

We utilize the following metrics to assess the model's performance on the MSA task: (1) Acc7: the accuracy for categorizing sentiment intensity into seven classes; (2) Acc2: the binary accuracy for distinguishing between non-negative (positive and neutral) and negative sentiments; (3) F1 score: a metric that measures the balance between precision and recall of binary sentiment classification; (4) MAE: the mean absolute error between model predictions and annotated labels; and (5) Corr: the correlation coefficient reflecting the strength and direction of the relationship between the model's predictions and the annotators' assessments. For determining Acc7, the model's predictions are rounded to the nearest integer within the range from -3 to 3. It is important to note that for calculating Acc2 and the F1 score, sentiment intensities are binned into two categories: non-negative (comprising positive and neutral sentiments) and negative.

For the MER task, following prior work [19], we calculate the accuracy and weighted F1 score of the model for recognizing emotions.

E. Implementation Details

The proposed MGT is implemented and trained using PyTorch, leveraging CUDA 11.6 and PyTorch 1.7.1, with the computations being performed on the RTX2080Ti GPU. The parameters of MGT are optimized by AdamW [78] optimizer. For specifics on the hyperparameter configurations, please refer to Table I.

¹<https://github.com/librosa/librosa>

TABLE I
HYPERPARAMETERS OF MGT

Batch Size n	100
Learning Rate for T5	2e-5
Learning Rate for Other Modules	8e-4
Weight Decay	5e-5
Training Epochs	30
Weight of Losses α	0.1
Margin γ	0.2

TABLE II
THE COMPARISON WITH BASELINES ON CMU-MOSI

	Acc7 ($\uparrow$)	Acc2 ($\uparrow$)	F1 ($\uparrow$)	MAE ($\downarrow$)	Corr ($\uparrow$)
LF-DNN [79]	33.6	78.0	77.9	0.978	0.658
Graph-MFN [11]	34.4	77.9	77.8	0.939	0.656
MFM [80]	33.3	77.7	77.7	0.948	0.664
MISA [34]	43.5	81.8	81.7	0.752	0.784
Self-MM [66]	45.8	82.7	82.6	0.731	0.785
MMIM [69]	45.0	83.0	82.9	0.738	0.781
DMD [72]	46.4	84.2	84.1	0.709	0.796
ITHP [36]	47.7	86.1	86.1	0.663	0.856
MCL-MCF [†] [27]	-	84.9	84.7	0.692	0.799
ConFEDE [†] [49]	42.27	84.17	84.13	0.742	0.782
CLGSI [†] [26]	47.96	83.97	83.63	0.703	0.790
UniMSE [†] [19]	48.68	85.85	85.83	0.691	0.809
MGT	50.44	86.30	86.26	0.659	0.822

The results of the baselines denoted with $\dagger$ are directly sourced from their respective publications, and other results are obtained under the same environment as ours. When calculating Acc2 and F1 score, we classify nonnegative/ negative utterances, with non-negative utterances encompassing both positive and neutral utterances.

TABLE III
THE COMPARISON WITH BASELINES ON CMU-MOSEI

	Acc7 ($\uparrow$)	Acc2 ($\uparrow$)	F1 ($\uparrow$)	MAE ($\downarrow$)	Corr ($\uparrow$)
LF-DNN [79]	52.1	78.6	79.0	0.561	0.723
Graph-MFN [11]	51.9	81.9	82.1	0.569	0.725
MISA [34]	52.2	81.6	82.0	0.550	0.758
MFM [80]	50.8	80.3	80.7	0.580	0.722
Self-MM [66]	53.0	82.6	82.8	0.540	0.763
MMIM [69]	53.1	81.9	82.8	0.547	0.752
DMD [72]	54.3	84.6	84.6	0.531	0.759
ITHP [36]	52.2	85.4	85.4	0.550	0.792
MCL-MCF [†] [27]	-	84.2	84.4	0.536	0.767
ConFEDE [†] [49]	54.86	81.65	82.17	0.522	0.780
CLGSI [†] [26]	<u>54.56</u>	84.01	84.21	0.532	0.763
UniMSE [†] [19]	54.39	<u>85.86</u>	<u>85.79</u>	<u>0.523</u>	0.773
MGT	54.30	86.05	86.09	0.522	0.764

F. the Comparison With Baselines

Results on MSA: In this section, we evaluate the performance of the proposed MGT against other well-known models on the MSA task. The comparative results on CMU-MOSI and CMU-MOSEI datasets are presented in Tables II and III, respectively. For each evaluation metric, the top-performing result is emphasized in bold, while the runner-up is indicated with underlines. It can be inferred from Tables II and III that MGT obtains the best results in the majority of metrics on CMU-MOSI and CMU-MOSEI datasets. To be more specific, compared to our closest baseline UniMSE [19] that injects multimodal fusion layer after attention layer, MGT outperforms it in all the metrics on CMU-MOSI. On CMU-MOSEI, MGT outperforms UniMSE in Acc2, F1 score and MAE, and is inferior to it in terms of Acc7 and Corr. MGT also outperforms contrastive learning based methods ConFEDE [49], MCL-MCF [27], and CLGSI [26] by a considerable margin. Moreover, MGT outperforms a strong baseline ITHP [36] that applies the powerful DeBERTa [81] as language network in the majority of evaluation

TABLE IV
THE COMPARISON WITH BASELINES ON MELD

Model	Accuracy	F1 score
LMF [71]	61.15	58.30
TFN [8]	60.70	57.74
MFM [80]	60.80	57.80
DialogueGCN [76]	59.46	58.10
DAG-ERC [75]	-	63.65
MMGCN [74]	-	58.65
MM-DFN [73]	62.49	59.46
UniMSE [19]	<u>65.09</u>	65.51
MGT	65.82	<u>63.78</u>

TABLE V
ABLATION STUDIES ON THE CMU-MOSI. ‘MPL’ DENOTES THE MULTIMODAL PROMPT LEARNING

	Acc7 ($\uparrow$)	Acc2 ($\uparrow$)	F1 ($\uparrow$)	MAE ($\downarrow$)	Corr ($\uparrow$)
W/O Margin Loss	47.23	86.15	86.14	0.685	0.816
W/O Matching Loss	48.98	85.86	85.72	0.648	0.829
W/O Margin Loss and Matching Loss	46.79	85.86	85.78	0.753	0.802
W/O Multimodal Flow Module	42.27	85.13	85.08	0.772	0.774
W/O Gating Layer	46.79	85.28	85.19	0.684	0.810
W/O Task Prompt	49.42	85.86	85.83	0.683	0.817
MPL1	49.27	85.71	85.66	0.655	0.820
MPL2	49.56	86.15	86.09	0.670	0.827
MGT	50.44	86.30	86.26	0.659	0.822

In MPL2, we set 16 trainable prompt vectors, which performs well in our experiment.

metrics. Combining the results on two datasets, MGT reaches the best performance and outperforms competitive baselines ITHP, UniMSE, ConFEDE, MCL-MCF, and CLGSI. The improvement of MGT is noteworthy, especially considering that the state-of-the-art approaches have already suppressed human performance [8]. We argue that it is because compared to other methods, we design a parallel multimodal flow module to identify the discriminative level of multimodal representations and explicitly control the flow of nonverbal modalities via the gating mechanism. Moreover, we propose margin and matching losses to maintain the unique information of each modality during the modality alignment process, which address the shortcomings of cross-modal contrastive learning [19], [26], [27], [49].

Results on MER: In addition, to assess the generalization capability of MGT across various multimodal tasks, we conduct experiments to compare MGT with other models on the MER task. We can infer from Table IV that compared to the state-of-the-art models, the proposed MGT reaches the highest score in Accuracy and the second-highest in the F1 score. Overall, MGT delivers performance that is on a par with the state-of-the-art models on the MELD dataset, showcasing the efficacy and broad applicability of the proposed approach.

G. Ablation and Comparison Experiments

In this section, we conduct ablation studies to assess the contribution of each individual component within the MGT framework, and the results are detailed in Table V.

1) The importance of margin loss and matching loss: As we can infer from the ‘W/O Margin Loss’, ‘W/O Matching Loss’, ‘W/O Margin Loss + Matching Loss’ cases in Table V, when either of the margin loss and matching loss is removed, the performance of the model drops by a considerable margin. Moreover, when both losses are removed, a more significant decline in performance is observed (especially in MAE). These

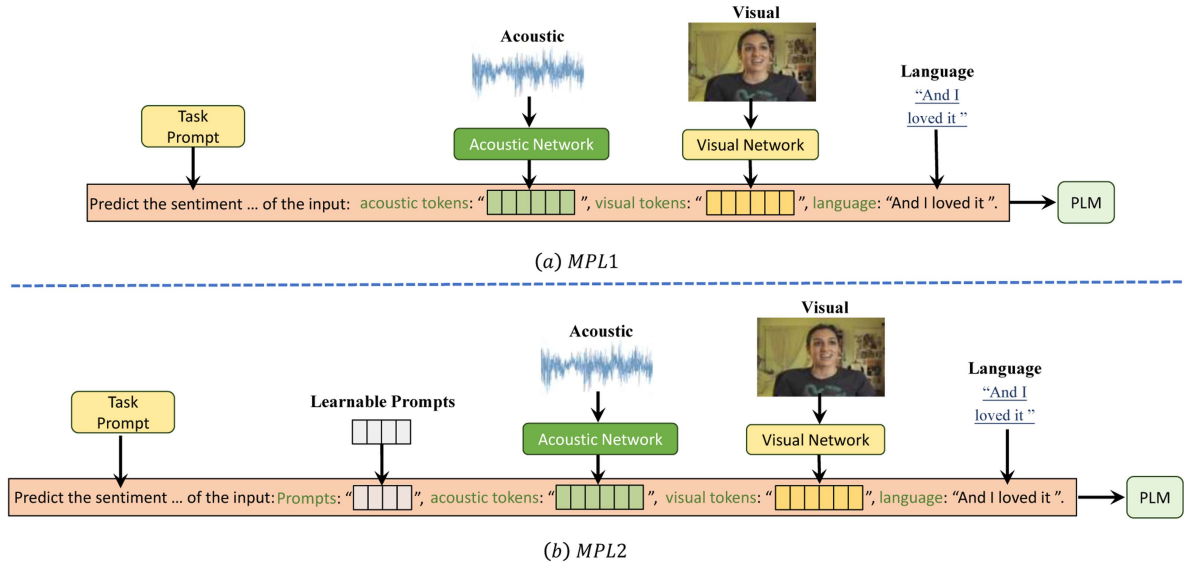


Fig. 5. The prompt learning techniques for comparison. The visual/acoustic network is the defaulted LSTM network.

results demonstrate the importance of the distribution alignment losses to force the distributions of nonverbal modalities to be similar to the distribution of language modality, such that the pre-trained language model can better understand the features from nonverbal modalities.

2) *The importance of multimodal flow module:* In the case of ‘W/O Multimodal Flow Module’, we remove the multimodal flow module such that the model relies on the language modality to generate the results. It is evident that there is a significant degradation in performance across all the evaluation metrics (about 8 points in Acc7 and the performance in MAE and Corr also suffers from significant drops), indicating that it is important to leverage nonverbal modalities to improve the understanding of the human multimodal language.

3) *The importance of gating layer:* In the case of ‘W/O Gating Layer’, we remove the gating layer in the multimodal flow module such that the multimodal representation output by the cross-modal additive attention layer is directly added with the original language features. We notice that there is a decline in performance by approximately 3.7 points in Acc7 and 1.0 points in Acc2, with similar reductions observed in other evaluation metrics. While the decrease in performance is not as pronounced as when the multimodal flow module is removed, these outcomes suggest that identifying the discriminative level and controlling the flow of the multimodal representation helps to improve the performance of human multimodal language understanding.

4) *Discussion on task-specific prompts:* As for the case of ‘W/O Task Prompt’ where we remove the task-aware prompts that concatenate with the language input, we can infer from Table V that the model’s performance experiences a decrease of roughly 1.0 points in the Acc7 metric and 0.4 points in the Acc2 metric, with the performance on other metrics also showing a minor reduction. Actually, MGT still reaches relatively good performance without task-aware prompts. Nevertheless, the minor decline in results highlights the beneficial

role of task-aware prompts to reduce the difficulty of model learning and task comprehension.

5) *Discussion on multimodal prompt learning:* In this section, we compare the proposed multimodal flow module with popular multimodal prompt learning strategies [60], [62], [63], [64], [65] by removing multimodal flow module and injecting nonverbal features via the prompt learning pipeline. Specifically, we investigate two different types of prompt learning: 1) MPL1: It concatenates nonverbal features with language tokens, and uses the task prompt defined in Section III-D to guide the PLM to infer the sentiment according to the multimodal input (see Fig. 5(a)); and 2) MPL2: It adds learnable prompts to MPL1 that further effectively adapt the PLM to the MSA task (see Fig. 5(b)). As shown in the cases of ‘MPL1’ and ‘MPL2’ in Table V, the two prompt learning techniques both achieve competitive results, and ‘MPL2’ slightly outperforms ‘MPL1’ combining all the evaluation metrics. Nevertheless, both of the MPL versions are inferior to the proposed multimodal flow module in terms of the performance on MSA. We argue that this is because the proposed parallel multimodal flow module explicitly controls the influence of non-discriminative multimodal information and enables the dominant role of language modality, which is of significance because nonverbal features often contain noisy information in the context of MSA.

H. Discussion on Modality Alignment

We perform comparison experiments to replace the proposed distribution alignment losses with other existing approaches to evaluate the advantages of the proposed losses over the existing approaches. Specifically, we replace the margin loss and matching loss with the widely-used cross-modal contrastive learning loss [19], [25], which can align the distributions of different modalities. As shown in Table VI, when the margin and matching losses are replaced with the contrastive loss, the

TABLE VI
COMPARISON EXPERIMENTS ON CMU-MOSI

	Acc7 (↑)	Acc2 (↑)	F1 (↑)	MAE (↓)	Corr (↑)
Contrastive Learning	47.67	85.57	85.53	0.696	0.807
MSE	48.98	86.01	85.92	0.659	0.820
Hard Triplet Loss	49.56	86.44	86.39	0.678	0.818
Margin Loss ($\gamma = 0$)	47.81	85.86	85.84	0.682	0.818
Margin Loss ($\gamma = 0.1$)	48.54	86.01	85.95	0.663	0.827
Margin Loss ($\gamma = 0.5$)	50.15	86.30	86.24	0.663	0.820
Margin Loss ($\gamma = 1$)	49.56	85.86	85.84	0.664	0.812
MGT ($\gamma = 0.2$)	50.44	86.30	86.26	0.659	0.822

model's performance is dropped by approximately 2.8 points for the Acc7 metric and 0.7 points for the Acc2 metric, with a corresponding downturn observed in the scores for other metrics as well. This is reasonable because we explicitly align the distributions of different modalities and also seek to preserve modality-specific information, such that we can obtain more expressive and powerful multimodal representation.

Moreover, we seek to replace the margin loss with hard triplet loss and mean square error (MSE), which are similar to our margin loss. From the results in Table VI, we can infer that when the margin loss is replaced with MSE, the performance drops by a slight margin. We argue that this is because MSE explicitly demands that features of various modalities from the same sample match precisely, potentially leading to a loss of information that is specific to each modality. Moreover, MSE does not sample negative pairs for training. For hard triplet loss, the anchor is language representation, the positive is the corresponding visual/acoustic representation from the same sample, and the negative is set to the visual/acoustic representation from other sample in the mini-batch that shares the highest similarity with the anchor. The results in Table VI indicate that hard triplet loss reaches the best results in Acc2 and F1 score (the margins of hard triplet loss and margin loss are the same). Nonetheless, synthesizing the results across all metrics, our margin loss still slightly surpasses hard triplet loss. The excellent performance of hard triplet loss results in that it seeks to maintain the unique characteristics of individual modalities while aligning them and considers the hardest negative sample for better alignment. And our margin loss slightly outperforms hard triplet loss mainly because margin loss considers multiple negative samples instead of only one negative sample.

Finally, we further explore the variability in model performance in response to adjustments in the value of the margin γ in the margin loss. When γ is set to zero, the proposed MGT exhibits its poorest performance, which is reasonable because the modality alignment is very weak when γ is zero. The results indicate that γ should be a moderate number to reach a better performance. Generally, our MGT is capable of achieving satisfactory performance across a broad spectrum of hyperparameter configurations for γ , which to some degree implies the robustness of the proposed margin loss.

I. Visualization of Embedding Space

We present a t-SNE [82] visualization of unimodal representations in the embedding space to evaluate the efficacy of the proposed matching and margin losses. In line with prior studies [34], [83], we conduct this visualization by projecting

TABLE VII
THE RESULTS OF UNIMODAL AND BIMODAL COUNTERPARTS OF MGT ON CMU-MOSI

	Acc7 (↑)	Acc2 (↑)	F1 (↑)	MAE (↓)	Corr (↑)
Language	42.27	85.13	85.08	0.772	0.774
Acoustic	15.74	53.21	53.38	1.472	0.076
Visual	16.91	50.15	50.19	1.491	0.034
Language-Visual	49.42	86.44	86.38	0.665	0.801
Language-Acoustic	49.56	86.15	86.11	0.667	0.820
Visual-Acoustic	17.49	53.79	53.84	0.137	0.074
MGT	50.44	86.30	86.26	0.659	0.822

each unimodal representation into a two-dimensional feature space via the t-SNE algorithm [82], for the reason that t-SNE can discern the local structure of high-dimensional features. The results are depicted in Fig. 6, showcasing the visualizations of unimodal representations with/without margin and matching losses. We can infer from Fig. 6(a) that, without alignment losses, the data points from different modalities are pushed away and are relatively dispersed in the feature space. When matching loss is utilized (see Fig. 6(b)), the data points of three modalities are uniformly distributed and more compact in the embedding space, and their distribution gap is reduced. And when the margin loss is applied (see Fig. 6(c)), there are some overlaps of the data points from different modalities, especially for the language and acoustic data points. The differences of visualizations between margin loss and matching loss mainly lie in that the matching loss aligns the modalities from a global perspective to match the means and variances of features from different modalities, which does not require the features of each specific sample to be aligned. In contrast, the margin loss explicitly aligns the features of different modalities in a sample-specific manner, which is more likely to result in the mixture of features from different modalities of certain samples. The visualizations also suggest that the language and acoustic modalities are easier to align than the language and visual modalities. This is mainly because language and acoustic sequences are extracted from the same speech signal.

Despite the inevitable existence of distribution gap due to the heterogeneous nature of unimodal representations, MGT can effectively reduce the gap between them and encourage the distributions of nonverbal modalities to be similar to the distributions of language modality. Notably, we can notice that although the distribution gap between modalities is reduced after applying the proposed distribution alignment losses, their gap still exists. It is because different modalities come from heterogeneous data sources and their features are extracted using tools with significantly varying characteristics, and thus it is very difficult to explicitly align the distributions of various modalities.

J. Discussion on Unimodal and Bimodal Models

In this section, we present the results of unimodal and bimodal counterparts of MGT on the CMU-MOSI and CMU-MOSEI datasets. Observing the results in Tables VII and VIII, we can draw the subsequent conclusions:

1) *The effectiveness of individual modalities:* The language modality, leveraging the powerful T5 framework [14] as its unimodal network, demonstrates a marked improvement in

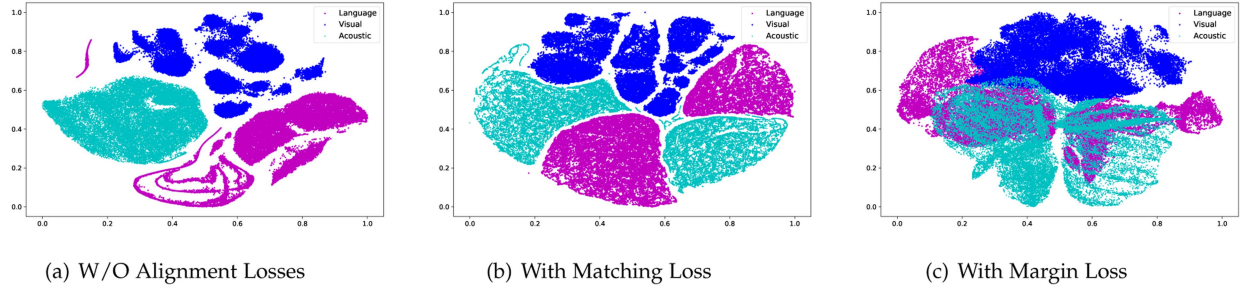


Fig. 6. T-SNE visualization of unimodal representations. ‘W/O Alignment Losses’ denotes the matching loss and margin loss are removed.

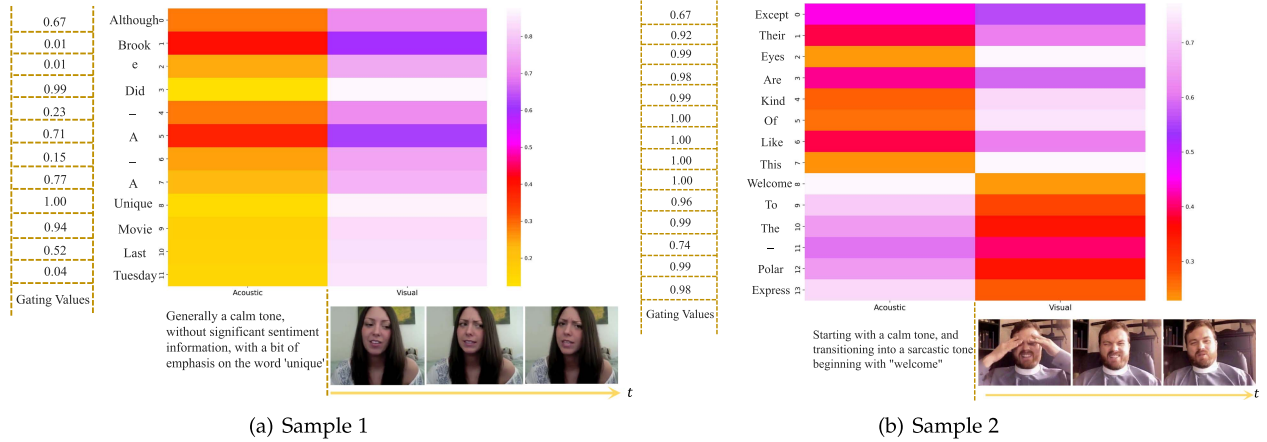


Fig. 7. Case Analysis for Multimodal Sentiment. We present two samples from CMU-MOSI testing set. ‘_’ is a special token in T5 tokenizer that corresponds to ID 3.

TABLE VIII
THE RESULTS OF UNIMODAL AND BIMODAL COUNTERPARTS OF MGT ON CMU-MOSEI

	Acc7 (↑)	Acc2 (↑)	F1 (↑)	MAE (↓)	Corr (↑)
Language	51.77	84.78	84.53	0.549	0.741
Acoustic	39.71	64.61	61.48	1.052	0.076
Visual	38.42	62.46	62.21	1.043	0.147
Language-Visual	52.92	85.92	85.94	0.534	0.755
Language-Acoustic	52.80	86.05	86.02	0.539	0.758
Visual-Acoustic	41.21	64.43	63.57	0.997	0.145
MGT	54.30	86.05	86.09	0.522	0.764

performance compared to both visual and acoustic modalities. Furthermore, combining the results on two datasets, the acoustic modality slightly outperforms the visual modality. Nonetheless, the performance of these two modalities remains suboptimal, which verifies our motivation that acoustic and visual modalities should serve as auxiliary modalities and additional design should be considered to prevent noisy information of nonverbal modalities from degrading the performance of PLM. A contributing factor for the poor performance of nonverbal modalities is the low quality and lack of sentiment-sensitivity in the extracted features of nonverbal modalities. This observation suggests a potential avenue for enhancing the multimodal sentiment analysis system: the extraction of high-quality sentiment-sensitive features from nonverbal modalities.

2) *The analysis on bimodal frameworks*: Overall, the performance of bimodal frameworks surpasses that of their unimodal

counterparts, and the trimodal system exhibits superior performance compared to bimodal systems. These findings indicate that incorporating additional modalities can enhance the performance of the framework.

K. Case Analysis for Multimodal Sentiment

In this section, we provide case studies of attention matrices and gating values to analyze whether the multimodal flow module can accurately leverage the useful information in visual and acoustic modalities. Note that here the attention weights for task prompt and special tokens are removed for clarity and simplicity.

We start with a simple case, in which the speaker’s spoken language is ‘Although Brooke did a a unique movie last Tuesday’. As demonstrated in Fig. 7(a), the speaker does not convey significant sentiment information via a calm tone (although she slightly emphasizes the word ‘unique’), but she begins to frown and display negative emotional cues starting with the word ‘did’. The model captures the negative sentiment conveyed by the visual modality and assigns larger weights to the correlations between language and visual information, especially focusing on the correlations of ‘did’ and ‘unique’ with visual features. The gating mechanism also assigns larger gating values to the time steps where ‘did’ and ‘unique’ are uttered. Using the proposed multimodal flow module, the proposed MGT accurately determines the sentiment score as weak negative. The visualization suggests that the cross-modal additive attention can

accurately identify important interactions between the language modality and other modalities, and the gating mechanism can selectively highlight discriminative information.

Moreover, we consider the challenging modality contradictory problem, where the language modality expresses sentiment information that is contradictory to visual and acoustic modalities. As shown in Fig. 7(b), based on the spoken language, one can easily draw the conclusion that the speaker may express a slightly positive sentiment, especially considering the word ‘welcome’ is usually associated with positive sentiment. However, as can be seen from the figure, the speaker exhibits an expression of disgust at the beginning, and later adopts a sarcastic tone, which expresses negative sentiment. The proposed MGT can accurately identify important correlation between language and nonverbal modalities. At the beginning, the model correctly identifies the correlation between language and visual information as carrying a significant amount of emotional information, particularly focusing on the associations of ‘eyes’ and ‘this’ with visual features, assigning them larger weights. In this way, the model is able to determine the negative expressions carried by the speaker when uttering these words. As the speaker begins to use sarcastic and teasing tones to express negative sentiment, the model shifts its focus to the correlations between language and acoustic information, especially the associations of ‘welcome’ with acoustic features, enabling the model to accurately recognize the negative acoustic information carried by the speaker when saying ‘welcome’. Moreover, most of the gating values become close to 1, allowing the information from nonverbal modalities to pass into the main pipeline of PLM and compensate for the negative effect of language information. Using the designed multimodal flow module, the model correctly predicts the sentiment score as -0.93, which is close to the annotated label -0.6. And without the multimodal flow module, the prediction of the model becomes 0.37, highlighting the importance of nonverbal information for handling the modality contradictory problem. The visualization results suggest that although we treat the language modality as the dominant modality, the model still can focus on nonverbal features when the language modality conveys contradictory information to annotated labels.

L. Visualization of Gating Values

In this section, we provide a visualization for the gating values of multimodal representations to analyze whether the proposed method can identify the noise within nonverbal modalities. Specifically, we randomly select a batch of testing data for visualization, and then use Gaussian noise to replace visual and acoustic features to analyze whether the proposed gating mechanism can explicitly filter out noisy features.

It can be seen from Fig. 8 that the model can selectively highlight useful information in multimodal representations when nonverbal features are not entirely noisy. However, when visual and acoustic features are replaced with Gaussian noise, most of the gating values become close to zero, suggesting that the model can accurately identify the noise within nonverbal features and assign small gating values when nonverbal features are entirely

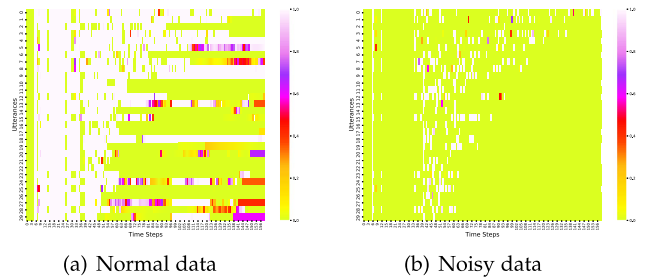


Fig. 8. Visualization of gating values.

TABLE IX
DISCUSSIONS OF DIFFERENT UNIMODAL NETWORKS ON CMU-MOSI DATASET

	Acc7 ($\uparrow$)	Acc2 ($\uparrow$)	F1 ($\uparrow$)	MAE ($\downarrow$)	Corr ($\uparrow$)
GRU	48.83	86.15	86.07	0.668	0.816
Transformer	50.44	86.44	86.41	0.645	0.828
Transformer-XL	50.15	86.59	86.56	0.643	0.828
LSTM (Default)	50.44	86.30	86.26	0.659	0.822

noisy. Interestingly, not all the gating values become zero, which is reasonable because multimodal representations also contain information from the language modality that improves the discriminative level of multimodal features.

M. Discussion on Unimodal Networks

To be consistent with our nearest baseline [19], we use LSTM networks as unimodal networks for acoustic and visual modalities. In this section, we change the unimodal networks of acoustic and visual modalities to investigate the effectiveness of injecting different kinds of nonverbal representations on the performance of PLM. Specifically, we evaluate the performance of GRU networks [84], Transformer [24], and Transformer-XL [85].

It can be seen from Table IX that GRU networks are slightly inferior to LSTM networks, which is reasonable because GRU can be regarded as a simpler version of LSTM and has fewer parameters. In contrast, when Transformer and Transformer-XL are used to extract nonverbal features, the performance of the model is further improved. We speculate that this is because Transformer and Transformer-XL are able to handle long sequences and prevent the forgetting of earlier historical information in the unimodal sequences in a more effective way via the attention mechanism that connects every two time steps. Moreover, attention mechanisms in Transformer variants can process the sequences in parallel at the time dimension, which can effectively reduce the training time. These results demonstrate the potential of the proposed MGT, indicating that the performance of our model can be further improved with more advanced acoustic and visual networks.

V. CONCLUSION

In this paper, we explore a novel solution that injects multimodal information into PLMs from a different perspective. Specifically, we propose a parallel multimodal flow module to process multimodal information to prevent noisy multimodal information from damaging the performance of PLM. In addition, we devise a gating mechanism to control the flow of nonverbal

information by estimating the discriminative information hidden in nonverbal modalities. Margin and matching losses are further designed to narrow the gap between various modalities, which can effectively preserve modality-specific information. In our framework, nonverbal modalities serve as auxiliary roles, providing the language model with additional information and improving the understanding of human multimodal language. Experiments verify that MGT reaches state-of-the-art performance on MSA and MER.

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